

Arithmetic applications of the categorical local Langlands correspondence

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Lecture 1. (G, X) Shimura datum, G conn. red. gr / $\mathbb{Q}$

$$X = G(\mathbb{R}) - \text{conj. class of } h: \text{Res}_{\mathbb{C}/\mathbb{R}} G_m \rightarrow G_{\mathbb{R}}$$

$$\mu: G_m \rightarrow G_m \xrightarrow{h} G_{\mathbb{R}} \text{ Hodge cocharacter}$$

$$\mathbb{E} \subset \mathbb{C} \text{ reflex field of } \langle \mu \rangle$$

$$\sim K \subset G(\mathbb{A}_f) \text{ open cpt} \sim \text{Sh}_K(G, X) \text{ quasi-proj. smooth var. / } \mathbb{E}$$

$$\text{Sh}_K(G, X)(\mathbb{C}) = G(\mathbb{Q}) \backslash X \times G(\mathbb{A}_f) / K$$

eg. $J = \begin{pmatrix} 0 & I_g \\ -I_g & 0 \end{pmatrix}$, $GSp_{2g} = \{ (g, c) \in GL_{2g} \times G_m : {}^t g J g = c J \}$

$$S^{\pm} = \{ Z = X + iY : Y \in M_{g \times g}(\mathbb{R}) : Z = {}^t Z, Y > 0 \text{ or } < 0 \}$$

$$(GSp_{2g}, S^{\pm}) \text{ Siegel Shimura datum}$$

$$\text{Sh}_K(GSp_{2g}, S^{\pm}) / \mathbb{Q} = \text{moduli of PPAV.}$$

$$(G, X) \text{ is of Hodge type if } (G, X) \hookrightarrow (GSp_{2g}, S^{\pm}).$$

Goal. Understand $R\Gamma_{\text{loc}}(\text{Sh}_{K^p}(G, X), \mathbb{Z}) \hookrightarrow G(\mathbb{Q}_p) \times W_{E^*} \times H_{K^p}$

with $G(\mathbb{Q}_p) \times W_{E^*} \times H_{K^p}$ -action, v p -adic place of E using CLLC.

Recall. Analytification $\text{Alg-stacks} / \mathbb{Z}_p \longrightarrow v\text{-stacks} / \mathbb{F}_p$ (v-sheafity)

$$X \longmapsto X^{\diamond}(R, R^+) = \{ (R^{\sharp}, R^{\sharp+}), \text{Spec } R^{\sharp+} \rightarrow X \}$$

$$X \longmapsto X^{\diamondsuit}(R, R^+) = \{ (R^{\sharp}, R^{\sharp+}), \text{Spec } R^{\sharp} \rightarrow X \}$$

$v\text{-stacks} \longrightarrow \text{perfect stacks}$

$$X \longmapsto X^{\text{red}}(R) = X(\text{Spd } R)$$

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Th (Zhang, Daniels - van Hatten - Kim - Zhang)

(1) (G, X) abelian type

$$\begin{array}{ccccc} \text{Sh}_{K^P}(G, X)^{\diamond, \diamond} & \subset & \text{Sh}_{K^P}(G, X)_{E_v}^{\diamond} & \xrightarrow{\pi_{HT}} & \text{Fl}_{\mu^{-1}}^{\diamond} \\ \downarrow & & \downarrow & \lrcorner & \downarrow \text{BL} \\ \text{Igs}_{K^P}^{\circ}(G, X) & \subset & \text{Igs}_{K^P}(G, X) & \xrightarrow{\bar{\pi}_{HT}} & \text{Bun}_G \end{array}$$

(2) (G, X) Hodge type, $K_P \subset G(\mathbb{Q}_p)$ parahoric, $K_P = G(\mathbb{Z}_p)$

$\mathcal{S}_{K^P K_P}(G, X)$ can. integral model $/\mathcal{O}_{E_v}$

$$\begin{array}{ccc} \mathcal{S}_{K^P K_P}(G, X)^{\diamond} & \xrightarrow{\pi_{\text{crys}}} & \text{Sh}_{G, \mu^{-1}}^{\text{an}} = \left\{ (S^{\#}, \Sigma; g\text{-torsn} / Y_S, \right. \\ & \lrcorner & \left. \pm \sigma^* \Sigma \xrightarrow{\frac{\Sigma \mu^{-1}}{SA}} \Sigma) \right\} \\ \downarrow & & \downarrow \\ \text{Igs}_{K^P}^{\circ}(G, X) & \xrightarrow{\bar{\pi}_{HT}} & \text{Bun}_G \end{array}$$

(3) Take $(-)^{\text{red}}$ to (2): $k = \bar{k}_v$

$$\begin{array}{ccc} \mathcal{S}_{K^P K_P}(G, X)_k^{\text{perf}} & \xrightarrow{\pi_{\text{crys}}} & \text{Sh}_{G, \mu^{-1}}^w \\ \downarrow & \lrcorner & \downarrow N+g \\ \text{Igs}_{K^P}^{\text{alg}}(G, X) & \xrightarrow{\bar{\pi}_{\text{crys}}} & \text{Isoc}_G \end{array}$$

Idea. (G, X) PEL type

$$\text{Igs}_{K^P}^{\circ}(G, X): (R, R^+) \mapsto \left\{ \text{ab. var.} / \text{Spa } R^+ / w \right\} / p\text{-quasi-irreg.}$$

$$\begin{array}{ccc} (2) (R, R^+) = (C, C^+) \text{ with unitt } (C^{\#}, C^{\#+}) & \left\{ \text{abel. var.} / C^{\#+} \right\} \rightarrow \left\{ p\text{-div. gp} / C^{\#+} \right\} \\ & \downarrow & \downarrow \\ & \left\{ \text{abel var.} / C^+ / \omega \simeq C^{\#+} / \omega^{\#} \right\} / p\text{-q.ing} & \rightarrow \left\{ p\text{-div gp} / C^+ / \omega \right\} / p\text{-q.ing} \\ & \text{Serre-Tate theory.} & \end{array}$$

$$\begin{array}{ccc}
 (3) & R \text{ perfect.} & \{ \text{abel var.} / R \} \longrightarrow \{ p\text{-div} / R \} \\
 & & \downarrow \qquad \qquad \downarrow \\
 & & \{ \text{abel var} / R \} / p\text{-isog} \longrightarrow \{ p\text{-div} / R \} / p\text{-isog}
 \end{array}$$

Rem. Zhang: (G, X) PEL type AC, also have Igusa stack diagrams for minimal compactifications.

generic fibre $\quad \overline{\pi}_{HT} : \text{Igs}_{K^p}^\circ \longrightarrow \overline{Bun_{G, \leq \mu-1}}$

Def. $\mathcal{F}^{an} := (\overline{\pi}_{HT})_* \Lambda \in D(Bun_{G, \leq \mu-1})$

$\mathcal{F}_c^{an} := (\overline{\pi}_{HT})_! \Lambda$

Th (DvHKZ) $\text{Igs}_{K^p}^\circ \xrightarrow{p} \text{Spd } \overline{\mathbb{F}_p}$ is coh. smooth of dim 0 with $p^! \Lambda \simeq \Lambda$

$$\Rightarrow D_{\text{verd}}(\mathcal{F}^{an}) = \mathcal{F}_c^{an}.$$

Recall $T_\mu : D(Bun_G) \longrightarrow D(Bun_G)^{BW_{E_v}} \quad i_1 : B\mathcal{L}(\mathcal{O}_p) \longrightarrow Bun_G$

Th (DvHKZ) $C = \widehat{E_v}, \quad R\Gamma(\text{Sh}_{K^p, C}, \Lambda) \simeq (i_1)^* T_\mu(\mathcal{F}^{an})(-\frac{d}{2})[-d]$

$$R\Gamma_c(\text{Sh}_{K^p, C}, \Lambda) \simeq (i_1)^* T_\mu(\mathcal{F}_c^{an})(-\frac{d}{2})[-d]$$

compatible with $\mathcal{L}(\mathcal{O}_p) \times W_{E_v} \times H_{K^p}$ -actions, $\quad \begin{matrix} \cup \\ W_{E_v} \end{matrix} \quad \begin{matrix} \cup \\ H_{K^p} \end{matrix}$

Proof

$$\begin{array}{ccccc}
 \text{Sh}_{K^p}^{\circ, \diamond} / \mathcal{L}(\mathcal{O}_p) & \xrightarrow{p} & \text{Fl}_{\mu-1}^\diamond / \mathcal{L}(\mathcal{O}_p) & \rightarrow & * / \mathcal{L}(\mathcal{O}_p) \\
 & & \downarrow & \lrcorner & \downarrow i_1 \\
 & & \text{Hck}_{\leq \mu} & \xrightarrow{h_2} & Bun_G \\
 & & \downarrow h_1 & & \\
 \text{Igs}_{K^p} & \longrightarrow & Bun_G & &
 \end{array}$$

$$(i_1^*) T_\mu(\mathcal{F}^{an}) \simeq i_1^* h_{2!} h_2^* (\overline{\pi}_{HT})_* \Lambda \not\simeq p_* \Lambda = R\Gamma(\text{Sh}_{K^p}^{\circ, \diamond}, \Lambda) \xrightarrow[\widehat{W_v}]{\text{Lan-Sum}} R\Gamma(\text{Sh}_{K^p, C}, \Lambda).$$

Wz Eichler-Shimura relation holds for $R\Gamma(Sh_{K^p/K^p}, c, \Lambda)$

$$\left(\begin{array}{l} \text{GL}_2: Fr^2 + p \cdot T_p \cdot Fr + p \cdot S_p = 0 \\ k_p \text{ parabolic} \end{array} \right)$$

Semi-properness

Th (D_{VHKZ}) If $Sh_{K^p/K^p}(G, X)$ proper, then $F^{an} \simeq F_c^{an} \in PD(Bun_{G, \leq \mu^{-1}})^0$

Idea. fibre of $\pi_{HT}: Ig_{K^p}^\bullet \rightarrow Bun_G \simeq (Ig_{K^p}^b)^0$

$$\Rightarrow (i_b^*)^* F^{an} \simeq R\Gamma(Ig_{K^p}^b, \Lambda) \in Rep(G_b(\mathbb{Q}_p))^{\leq \langle 2p, \nu_b \rangle}$$

Mao: $Ig_{K^p}^b$ affine of $\dim \langle 2p, \nu_b \rangle$

□

Rem, (araiam-Hamann-Zhang: for PEL type AC, construct

$F_{(c)}^{an} \in PD(Bun_{G, \leq \mu^{-1}})^0$ compute intersection coh. of $Sh_{K^p}^{min}$.

Special fibre $Sh_{v_{min}}^{tame}(Isoc_G)$

$$\text{central sheet: } Rep(\hat{G}) \xrightarrow{\mathbb{Z}} \mathbb{Z}(Sh_{v_{min}}(I^+ \backslash LG/I^+))$$

$\downarrow$
 $\vee$

$\mathbb{Z}(V) \rtimes (-): Sh_{v_{min}}(I^+ \backslash LG/I^+) \ni \cdot$ bilinear over $Sh_{v_{min}}(I^+ \backslash LG/I^+)$.

$$Sh_{v_{min}}^{tame}(Isoc_G) = Tr(Sh_{v_{min}}(I^+ \backslash LG/I^+), Sh_{v_{min}}(I^+ \backslash LG/I^+)^{\otimes})$$

$\hookrightarrow$
 $\mathbb{Z}(V)$

$$\hookrightarrow Sh_{v_{min}}(I^+ \backslash LG/I^+) \xrightarrow{\mathbb{Z}(V) \rtimes (-)} Sh_{v_{min}}(I^+ \backslash LG/I^+)$$

$$N \times S^1 \downarrow \quad \sim \quad \downarrow$$

$$Sh_{v_{min}}^{tame}(Isoc_G) \xrightarrow{TV} Sh_{v_{min}}^{tame}(Isoc_G)$$

$$S_{K^p K^p}^{\text{pert}} \longrightarrow \text{Sh}_{G, \mu^{-1}}^W$$

$$\begin{array}{ccc} \downarrow & & \downarrow N+g \\ \text{Igs}_{K^p}^{\text{alg}} & \xrightarrow{\overline{\pi}_{\text{cris}}} & \text{Isoc}_{G, \mu^{-1}} \end{array}$$

Def. $\mathcal{F}^{\text{alg}} := (\overline{\pi}_{\text{cris}})_b^* \omega_{\text{Igs}_{K^p}^{\text{alg}}}$ $(-)_b$: right adj. to $(-)^!$.

$$\mathcal{F}_c^{\text{alg}} := (\overline{\pi}_{\text{cris}})_b^* \omega_{\text{Igs}_{K^p}^{\text{alg}}}^{\text{can}}$$

$$\left(\mathbb{D}^{\text{Adm}} \text{ on } \text{Shv}(\text{Igs}_{K^p}^{\text{alg}}), \omega_{\text{Igs}_{K^p}^{\text{alg}}}^{\text{can}} = \text{ID}^{\text{Adm}}(\omega_{\text{Igs}_{K^p}^{\text{alg}}}) \right)$$

- $X \xrightarrow{f} \text{Spec } k$ ptp scheme, $f_*: \text{Shv}(X) \rightarrow \mathbb{D}(\Lambda)$, admits a cts
right adjoint $f^{\#}$

Def $\omega_X^{\text{can}} = f^{\#} \omega$.

$$\begin{aligned} \mathcal{F} \in \text{Shv}_c(X), \text{ then } \text{Hom}(\mathcal{F}, f^{\#} \omega) &= \text{Hom}(f_* \mathcal{F}, \omega) \\ &= R\Gamma(X, \mathcal{F})^{\vee} \\ &= R\Gamma_c(X, \text{ID}(\mathcal{F})) \end{aligned}$$

then $\text{ID}_{\text{Igs}_{K^p}^{\text{alg}}}^{\text{can}}$ is descent of $\omega_{S_{K^p K^p}^{\text{pert}}}^{\text{can}}$

$$\text{Can show } \text{ID}^{\text{Adm}}(\mathcal{F}^{\text{alg}}) \simeq \mathcal{F}_c^{\text{alg}} \text{ in } \text{Shv}(\text{Isoc}_G)$$

$\mathcal{F}^{\text{alg, tame}} :=$ image of $\mathcal{F}^{\text{alg}}$ along right adjoint of $\text{Shv}^{\text{tame}} \hookrightarrow \text{Shv}$

$$\mathcal{F}_c^{\text{alg, tame}} := \longleftarrow \mathcal{F}_c^{\text{alg}} \longrightarrow$$

Th. (Gleason - Y.-Zhu) $R\Gamma(\text{Sh}_{K^p}, \Lambda)^{\text{tame}}(\frac{d}{2})[d] \simeq i_2^! T_{\mu}(\mathcal{F}^{\text{alg, tame}})$

(G, X) Hodge type $R\Gamma_c(\text{Sh}_{K^p}, \Lambda)^{\text{tame}}(\frac{d}{2})[d] \simeq i_2^! T_{\mu}(\mathcal{F}_c^{\text{alg, tame}})$

G splits / unres. ext^r
or Op

Sketch $\text{Rep}^{\text{tame}}(G(\mathbb{Q}_p))$ is gen. by $c\text{-ind}_{G(\mathbb{Z}_p)}^{G(\mathbb{Q}_p)} \pi$.

$L(\mathbb{F}_p)$ Levi factor of $G(\mathbb{Z}_p)$, $\pi =$ Deligne-Lusztig induction of cofree mon.

tilting object $\mathcal{F} \in \text{Sh}_{\text{min}}(U_L \backslash L/U_L)$

$$U_L \backslash L/U_L \longleftarrow \frac{L}{\text{Ad}_\sigma U_L} \longrightarrow \frac{L}{\text{Ad}_\sigma L} = \text{IBL}(\mathbb{F}_p)$$

Need. $\text{Hom}\left((i_1)_* c\text{-ind}_{G(\mathbb{Z}_p)}^{G(\mathbb{Q}_p)} \pi, T_\mu(\mathcal{F}^{\text{alg, tame}})\right) \simeq R\Gamma(\text{Sh}_{K^p} G(\mathbb{Z}_p), \epsilon, L_\pi^\vee)$

$\text{Sh}_{K^p}^{\text{pert}} G(\mathbb{Z}_p) \xrightarrow{\pi_{\text{crys}}} \text{Sh}_{G, \mu-1}^{\text{tame}} = \frac{L_{G, W_p} \text{Ad}_\mu(\mu^{-1}) W_p}{\text{Ad}_\sigma I^+} \xrightarrow{\delta} I^+ \backslash L_{G, I^+}$

$\downarrow \quad \quad \quad \downarrow$

$I_{G, K^p}^{\text{alg}} \xrightarrow{\bar{\pi}_{\text{crys}}} \text{Isoc}_{G, \mu-1}$

L_π local system on $\text{Sh}_{K^p} G(\mathbb{Z}_p)$ assoc. to π .

$$T_{\mu^{-1}}\left((i_1)_* c\text{-ind}_{G(\mathbb{Z}_p)}^{G(\mathbb{Q}_p)} \pi\right) \simeq N_{I^+}^\vee \delta^! \left(\mathbb{Z}(V_{\mu^{-1}}) * \mathcal{F}\right)$$

$$\text{Hom}\left(N_{I^+}^\vee \delta^! \left(\mathbb{Z}(V_{\mu^{-1}})\right), \mathcal{F}^{\text{alg, tame}}\right)$$

$$\stackrel{!!}{(\bar{\pi}_{\text{crys}})_b \omega}$$

$$= \text{Hom}\left(\bar{\pi}_{\text{crys}}^! N_{I^+}^\vee \delta^! \left(\mathbb{Z}(V_{\mu^{-1}}) * \mathcal{F}\right), \omega_{I_{G, K^p}^{\text{alg}}}\right)$$

$$= \text{Hom}\left(\underbrace{\pi_{\text{crys}}^!}_{!!} \delta^! \left(\mathbb{Z}(V_{\mu^{-1}}) * \mathcal{F}\right), \omega_{\text{Sh}_{K^p}^{\text{pert}} G(\mathbb{Z}_p)}}_W\right)$$

$$\simeq R\Gamma\left(\text{Sh}_{K^p}^{\text{pert}} G(\mathbb{Z}_p), \text{ID}(W)\right).$$

Key $W = \mathbb{F}[I\pi][d]$.

$\quad \quad \quad \uparrow$ nearby cycle

Lecture 2. Over $\text{Spd } \mathcal{O}_C$, $C = \widehat{E}_V$.

$$\widetilde{\text{Sh}}_{g, \mu-1}^{\text{an}} = \left\{ (S^\sharp, \varepsilon, \sigma^* \varepsilon \rightarrow \varepsilon, (\varepsilon|_S) \times^{g_{\text{FP}}} L \right\}$$

u_L -reduction of

$(L/u_L)^{\diamond} \text{-fib.}$

$$\text{S}_{K^p G(\mathbb{Z}_p)}^{\diamond} \longrightarrow \text{Sh}_{g, \mu-1}^{\text{an}} = \left\{ (S^\sharp, \varepsilon \text{ } g\text{-torsion} / \gamma_S, \right. \\ \left. + \sigma^* \varepsilon \xrightarrow{\frac{\varepsilon_{\mu-1}}{S^\sharp}} \varepsilon) \right\}$$

Def. $\text{Sh}_{g, \mu-1}^{\text{res}} / \text{Spd } \mathcal{O}_C$ (Genestier-Lafforgue)

$$s \mapsto \left\{ (S^\sharp, \varepsilon_1, \varepsilon_2 \text{ } g\text{-torsion} / \widehat{S \cup S^\sharp}, \varepsilon_1|_{\widehat{S \cup S^\sharp} \setminus S^\sharp} \xrightarrow{\varepsilon_{\mu-1}} \varepsilon_2|_{\widehat{S \cup S^\sharp} \setminus S^\sharp}) \right\} \\ + \varepsilon_1|_{\widehat{S}} \simeq \sigma^* \varepsilon_2|_{\widehat{S}}$$

$$\widehat{S} = (\mathcal{O}_{Y_S})_{\widehat{S}} = w(R)$$

$$\widetilde{\text{Sh}}_{S, \mu-1}^{\text{an}} \longrightarrow \widetilde{\text{Sh}}_{g, \mu-1}^{\text{res}} \longrightarrow \text{Hck}_{I^+}^{\text{BD}}$$

$$\widehat{S^\sharp} = (\mathcal{O}_{Y_S})_{\widehat{S^\sharp}} = \text{BdR}^+(S^\sharp)$$

$$\text{S}_{K^p G(\mathbb{Z}_p)}^{\diamond} \longrightarrow \text{Sh}_{S, \mu-1}^{\text{an}} \longrightarrow \text{Sh}_{g, \mu-1}^{\text{res}}$$

$$\left\{ \begin{array}{l} S^\sharp, \varepsilon_1, \varepsilon_2 \text{ } I^+ \text{-torsion} \\ \widehat{S \cup S^\sharp} \\ + \varepsilon_1|_{\widehat{S \cup S^\sharp} \setminus S^\sharp} \simeq \varepsilon_2|_{\widehat{S \cup S^\sharp} \setminus S^\sharp} \end{array} \right\}$$

$/ \text{Spd } C$

$$(L/u_L)^{\diamond} \times^{g(\mathbb{Z}_p)} \text{Fl}_{\mu-1}^{\diamond} \longrightarrow \text{Hck}_{G, \mu-1} \times^{g(\mathbb{Z}_p)} (L/u)^{\diamond} \xrightarrow{\text{pro-unip} \times (I^+ | L_G / I^+)^{\circ}} \text{Hck}_{G, \mu-1}$$

$\downarrow$ $\text{propu} + T^{\diamond} \text{-torsion} \rightarrow$

$$\text{Sh}_{K^p, G(\mathbb{Z}_p)}^{\diamond, \diamond} \longrightarrow \underline{g(\mathbb{Z}_p)} \setminus \text{Fl}_{\mu-1}^{\diamond} \longrightarrow \text{Hck}_{G, \mu-1} \times \text{IB } \underline{g(\mathbb{Z}_p)}$$

$$\mathbb{Z}(\nu_{\mu-1})^* F = \mathbb{F}(S_{\mu-1} \otimes F)$$

Key input: $\text{S}_{K^p G(\mathbb{Z}_p)}^{\diamond} \longrightarrow \text{Sh}_{g, \mu-1}^{\text{res}}$ is "smooth".

$$\underline{\text{Th}} \text{ (Y.-Zhu)} \quad F^{\text{alg}} \in \text{Sh}_v(\text{Iso}_C)^{e, \geq 0}$$

$$F_C^{\text{alg}} \in \text{Sh}_v(\text{Iso}_C)^{e, \leq 0}$$

Sketch:

$$\pi_{\text{crys}} : Ig_{K^p}^{\text{alg}} \rightarrow Isoc$$

$$\text{fibre of } \pi_{\text{crys}} = Ig_{K^p}^b \xrightarrow{f} S_{K^p, I}^{\text{port}}$$

$$(i_b)! F_c^{\text{alg}} \simeq R\Gamma_c(S_{K^p, I}^{\text{port}}, t_x \Lambda) \simeq R\Gamma(Ig_{K^p}^{b, \min}, j_! \Lambda)$$

$$Ig_{K^p}^b \xrightarrow{f} S_{K^p, I}^{\text{port}}$$

$Ig_{K^p}^b$ is "well-positioned"

$$\begin{array}{ccc} j \downarrow & & \downarrow j \\ Ig_{K^p}^{b, \min} & \xrightarrow{f^{\min}} & S_{K^p, I}^{\text{port}, \min} \end{array}$$

$$j_! t_x \Lambda \simeq t_x^{\min} j_! \Lambda$$

$\hat{=}_{\text{Mao}}$

- $Ig_{K^p}^{b, \min}$ is affine of dim $\langle 2p, V_b \rangle$

- $Ig_{K^p}^b$ smooth of dim $\langle 2p, V_b \rangle$

Artin vanishing $\Rightarrow R\Gamma(Ig_{K^p}^{b, \min}, \Lambda) \in \text{Rep}(\hat{G}_b(\mathcal{O}_p))^{\leq \langle 2p, V_b \rangle} \Rightarrow F_c^{\text{alg}} \in \text{Shv}(Isoc_{\mathcal{G}})^{e, \leq 0} \square$

Comparison $\underline{\text{Th}}(GHIZ) \quad \Lambda \text{ torsion}, \psi: \text{Shv}(Isoc_{\mathcal{G}}) \xrightarrow{\sim} D(Bun_{\mathcal{G}})$

$$(i_2)! \longleftrightarrow (i_1)^*$$

$$\underline{\text{Th}}(GYZ) (1) \quad V \in \text{Rep}(\hat{G}), \quad \text{Shv}^{\text{tame}}(Isoc_{\mathcal{G}}) \xrightarrow{T_V} \text{Shv}^{\text{tame}}(Isoc_{\mathcal{G}})$$

$$\begin{array}{ccc} \psi \downarrow & \sim & \downarrow \psi \\ D(Bun_{\mathcal{G}}) & \xrightarrow{T_V} & D(Bun_{\mathcal{G}}) \end{array}$$

$$(2) \quad \psi(F^{\text{alg}}) \simeq F^{\text{an}}$$

$$\underline{\text{Cor.}} \quad F^{\text{an}} \in {}^p D(Bun_{\mathcal{G}})^{\geq 0}, \quad F_c^{\text{an}} \in {}^p D(Bun_{\mathcal{G}})^{\leq 0}.$$

Torsion vanishing $\Lambda = \bar{\mathbb{F}}_l, \bar{\mathcal{O}}_l$

$$\underline{\text{Th}}(GYZ) \quad l > h, \quad \Omega^{\text{tame}}: \text{Shv}^{\text{tame}}(Isoc_{\mathcal{G}}) \rightarrow \text{Ind Coh}_X(\text{Loc}_{\mathcal{G}}, \mathcal{O}_p^{\text{tame}})$$

c4 t-exact

exotic t-str.

c4 $\nearrow$

$\downarrow$ $\text{IndPerf}(\text{Loc}_{\mathcal{G}}, \mathcal{O}_p^{\text{tame}})$ good filter t-str.

$$\text{Cor A. } \text{IW}_\psi = (-\text{ind}_{\mathbb{I}^+}^{G(\mathbb{Q}_p)}) \psi, \quad \psi: \mathbb{I}^+ \rightarrow G(\mathbb{F}_p) \rightarrow \Lambda^\times$$

$G_{\mathbb{Q}_p}$ unram. (G, X) Hodge type

$$\text{Hom}(\text{IW}_\psi, R\Gamma(\text{Sh}_{K^p}, \Lambda)) \text{ is in deg } [d, 2d]$$

$$\text{Hom}(\text{IW}_\psi, R\Gamma_c(\text{Sh}_{K^p}, \Lambda)) \text{ is in deg } [0, d]$$

Cor. π irred depth 0 rep of $G(\mathbb{Q}_p)$ s.t. $\text{Hom}(\text{IW}_\psi, \pi) \neq 0$

then if π is a subquotient of $H^i(\text{Sh}_{K^p}, \Lambda)$, then $i \geq d$

— if — $H_c^i(\text{Sh}_{K^p}, \Lambda)$, then $i \leq d$

Rem. This is part of some conj. of Koshikane - Shin.

Proof of Cor A. $\text{Hom}(\text{IW}_\psi, R\Gamma(\text{Sh}_{K^p}, \Lambda))$

$$= \text{Hom}((i_1)_* \text{IW}_\psi, T_\mu(\mathcal{F}^{\text{als}, \text{tame}}))$$

$$\stackrel{\mathbb{L}_\psi^{\text{tame}}}{=} \text{Hom}_{\text{IndCoh}_K}(\text{Loc}_{\mathbb{L}_G, \mathbb{Q}_p}^{\text{tame}})(\mathcal{O}, V_\mu \otimes \mathbb{L}_\psi^{\text{tame}}(\mathcal{F}^{\text{als}, \text{tame}}))$$

$$= R\Gamma(\text{Loc}_{\mathbb{L}_G, \mathbb{Q}_p}^{\text{tame}}, \underbrace{V_\mu \otimes \mathbb{L}_\psi^{\text{tame}}(\mathcal{F}^{\text{als}, \text{tame}})}_{\text{in deg } \geq 0}) \text{ in deg } \geq 0. \quad \square$$

Define $\varphi: W_{\mathbb{Q}_p} \rightarrow {}^L G(\Lambda)$ s.s. L -par. with cuspidal support

$$\varphi_\mu: W_{\mathbb{Q}_p} \xrightarrow{\text{s.c.}} {}^L M(\Lambda) \quad {}^L M \subset {}^L G \quad \text{Levi}$$

$$\begin{array}{c} \searrow \varphi \\ \downarrow \\ {}^L G(\Lambda) \end{array}$$

φ is generic (weakly Langlands - Shahidi type) if $H^2(W_{\mathbb{Q}_p}, (\hat{\mathfrak{g}}/\hat{\mathfrak{m}})_{\varphi_\mu}) = 0$

Ex. $\varphi: W_{\mathbb{Q}_p} \rightarrow G_{\text{Lan}}(\Lambda)$, $\varphi = \varphi_1 \oplus \dots \oplus \varphi_n$, φ_i unram. char. $\hat{\mathfrak{g}}/\hat{\mathfrak{m}} \hookrightarrow {}^L M \xleftarrow{\varphi_\mu} W_{\mathbb{Q}_p}$

φ generic $\Leftrightarrow \varphi_i \neq \varphi_j(1)$, $\forall i \neq j$
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$$\begin{array}{ccc} \text{Loc}_{L_H, \mathbb{A}_p}^{\text{gen}} & \subset & \text{Loc}_{L_H, \mathbb{A}_p} \\ \downarrow & & \downarrow \\ \text{Spec } \mathbb{Z}_{L_H, \mathbb{A}_p}^{\text{gen}} & \subset & \text{Spec } \mathbb{Z}_{L_H, \mathbb{A}_p} \end{array}$$

Prop. $\mathcal{N}|_{\text{Loc}_{L_H, \mathbb{A}_p}^{\text{gen}}} \simeq \{0\}|_{\text{Loc}_{L_H, \mathbb{A}_p}^{\text{gen}}}$ in $\text{Sing}(\text{Loc}_{L_H, \mathbb{A}_p}^{\text{gen}})$.

Rem. If $\Lambda = \overline{\mathbb{A}_k}$, $\text{Loc}_{L_H, \mathbb{A}_p}^{\text{gen}}$ is smooth.

If Λ modular, $\text{Loc}_{L_H, \mathbb{A}_p}$ non-reduced.

Cor. $\text{IndCoh}_N(\text{Loc}_{L_H, \mathbb{A}_p}^{\text{gen}}) \simeq \text{IndPort}(\text{Loc}_{L_H, \mathbb{A}_p}^{\text{gen}})$

Cor. $\mathbb{L}_\psi^{\text{tame, gen}}: \text{Sh}_V^{\text{tame}}(\text{Isoc}_G)^{\text{gen}} := \text{Sh}_V^{\text{tame}}(\text{Isoc}_G) \otimes_{\mathcal{O}(\text{Loc}(\text{Spec } \mathbb{Z}^{\text{tame, gen}}))} \mathcal{O}(\text{Loc}(\text{Spec } \mathbb{Z}^{\text{tame}}))$

$$\begin{array}{ccc} & \searrow c_\psi & \\ \downarrow & & \\ \text{IndCoh}_N(\text{Loc}_{L_H, \mathbb{A}_p}^{\text{tame, gen}}) & \simeq & \text{IndPort}(\text{Loc}_{L_H, \mathbb{A}_p}^{\text{tame, gen}}) \end{array}$$

c_ψ is t-exact $\Rightarrow \mathbb{L}_\psi^{\text{tame, gen}}$ is t-exact

If V tilting $\hat{G}$ -rep, then $T_V: \text{Sh}_V^{\text{tame}}(\text{Isoc}_G)^{\text{gen}} \rightarrow \text{Sh}_V^{\text{tame}}(\text{Isoc}_G)^{\text{gen}}$ is t-exact.

Cor. $R\Gamma(\text{Sh}_{Kp}, \Lambda)^{\text{tame}} \otimes_{\mathbb{Z}^{\text{tame}}} \mathbb{Z}^{\text{tame, gen}}$ in degrees $[d, 2d]$

$R\Gamma_c(\text{Sh}_{Kp}, \Lambda)^{\text{tame}} \otimes_{\mathbb{Z}^{\text{tame}}} \mathbb{Z}^{\text{tame, gen}}$ in degrees $[0, d]$

Rem. Laraiani-Scholze, Koshikawa, Hamann-Lee, Y.-Zhu.
Principal series
parahoric

